

Let's Do Our Homework with AI!

AI for teachers, Hand on workshop

September 17, 2026

#1. Problem

In a magical isosceles triangle $\triangle ABC$ we have $|AC| = |BC|$. Let D be the midpoint between A and B . The inscribed circle of ABC intersects the line segment CD in a point E that is in the interior of the triangle. Suppose that $|AB| = 15$ and $|CE| = 8$. Determine $|AC|$.

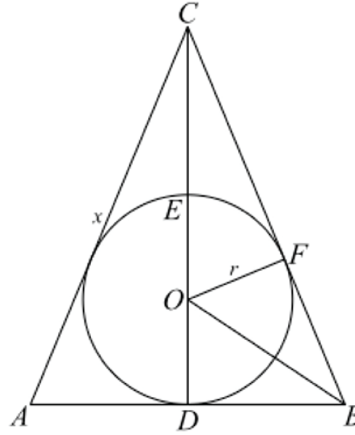


Figure 1: System operational scenario.

#2. Problem

For a given positive real number r and real x, y let $N_r(x, y)$ be the number of pairs of integers (m, n) satisfying $(x - m)^2 + (y - n)^2 \leq r^2$. Evaluate

$$\int_0^1 \int_0^1 N_r(x, y) dx dy$$

as a function of r .

#3. Problem

A collection of H-like atoms has some atoms in the lowest energy level 'A' and some atoms in a particular upper energy level 'B' and there are no atoms in any other energy level. The atoms of gas make transition to a higher energy level by absorbing mono-chromatic light of photon of energy 2.55 eV. Subsequent the atoms emit radiation of six different wavelengths. Some of the emitted photons have energy 2.55 eV and some have more and some have less than 2.55 eV.

- (1) Find the quantum number of initially excited level.
- (2) Find the atomic number of gas.
- (3) Find the ionisation energy for the gas atoms.
- (4) Find the maximum and minimum energies of the emitted photons.

#4. Problem

The Doppler flow meter is a particularly interesting medical application of the Doppler effect. This device measures the speed of blood flow, using transmitting and receiving elements that are placed directly on the skin, as in Figure. The transmitter emits a continuous sound whose frequency is typically about 5 MHz. When the sound is reflected from the red blood cells, its frequency is changed in a kind of Doppler effect because the cells are moving with the same velocity as the blood. The receiving element detects the reflected sound, and an electronic counter measures its frequency, which is Doppler-shifted relative to the transmitter frequency. From the change in frequency the speed of the blood flow can be determined. Typically, the change in frequency is around 600 Hz for flow speeds of about 0.1 m/s.

(1) Assume that the red blood cell is directly moving away from the source and the receiver. What is the (approx) speed of the sound wave in the blood?

- (A) 1700 m/s (B) 330 m/s (C) 5000 m/s (D) 3000 m/s

(2) An abnormal segment of the artery is narrowed down by an arteriosclerotic plaque to one-fourth the normal cross-sectional area. What will be the change in frequency due to reflection from the red blood cell in that region?

- (A) 150 Hz (B) 300 Hz (C) 600 Hz (D) 2400 Hz

(3) At what extra rate does the heart have to work due to this narrowing down of the artery? Assume the density to be 1.5 gm/cc and the area of the normal artery to be 0.1 cm^2 .

- (A) $1.125 \times 10^{-4} \text{ W}$ (B) $2.5 \times 10^{-4} \text{ W}$ (C) $6.25 \times 10^{-5} \text{ W}$ (D) $5.625 \times 10^5 \text{ W}$

#5. Problem

Write an essay of about 400 words, discussing whether AI is a tool or an agent.

#6. Problem

Translate the paragraph below into your native language.

But does it really matter what I talk about this morning? Because, as we all know, most addresses are soon forgotten. As Abraham Lincoln said in his Gettysburg Address, "You will little note nor long remember what I say here." And yes, Lincoln was right. No one remembers the Gettysburg Address. So, too, will my words likely be forgotten.

#7. Problem

Design an illustrated step-by-step guide for the recipe of sushi in a poster format.